

Merehan Ibraheem Elmotasem Hassan

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SUMMARY

Computer Science and Artificial Intelligence undergraduate at Cairo University with hands-on experience in Machine Learning, Deep Learning, NLP, and Data Engineering. Passionate about building AI applications using LLMs, RAG, and modern ML frameworks, with experience developing end-to-end AI solutions on real-world datasets.

EXPERIENCE

NLP Trainee

National Telecommunication Institute (NTI)

Jul 2026 – Aug 2026

- Completed an intensive one-month training program focused on Natural Language Processing (NLP).
- Worked on NLP fundamentals including text preprocessing, feature extraction, word embeddings, and deep learning approaches for text classification.
- Built practical NLP applications using Python and modern NLP libraries.

Data Engineering Trainee

Digital Egypt Builders Initiative (DEBI)

Jul 2025 – Dec 2025

- Completed an intensive training program in Data Engineering
- Worked on hands-on labs and real-world case studies using Python, SQL, and modern data processing tools.
- Gained practical experience in building data pipelines and working with databases.

Summer Intern – Banking Operations

Arab African International Bank (AAIB)

Jul 2025 – Aug 2025

- Exposure to digital banking services and internal operations.
- Supported branch operations and customer assistance.
- Promoted youth banking programs and learned customer interaction practices.

PROJECTS

AI Research Paper Assistant – NTI Graduation Project

Team Project

- Developed an AI-powered research paper assistant using LLMs and Retrieval-Augmented Generation (RAG).
- Built a pipeline for PDF processing, semantic search, and context-aware question answering.
- Generated summaries, extracted key contributions, and explained technical concepts using Qwen 2.5 Instruct.
- **Technologies:** Python, Streamlit, LangChain, FAISS, PyMuPDF, Ollama, Qwen 2.5 Instruct.
- GitHub: Project Repository

NLP Sentiment Analysis (Machine Learning, Deep Learning) (Team Project)

- Built an end-to-end NLP pipeline for sentiment classification using ML and deep learning models.
- Performed text preprocessing and feature engineering including tokenization, lemmatization, stop-word removal, and TF-IDF/sequence preparation.
- Applied FastText embeddings and developed LSTM-based models for classification.
- Evaluated models using Accuracy, Precision, Recall, and F1-score.
- GitHub: Project Repository

FinTech Credit Scoring Pipeline – DEPI Graduation Project (Team Project)

- Developed an end-to-end data engineering pipeline using Medallion architecture on Azure (Bronze, Silver, Gold layers).
- Built ETL workflows using Azure Data Factory and Databricks (PySpark) for large-scale financial data processing.
- Implemented ML models (Logistic Regression, Gradient Boosting) for credit risk classification (Low, Medium, High).
- Designed Power BI dashboards for risk analysis and business insights.

Deep Learning for Image Classification & Transfer Learning (Team Project)

- Built CNN-based image classification models using CIFAR-10 dataset with data augmentation and regularization techniques.
- Experimented with preprocessing methods including normalization and standardization.
- Applied transfer learning using pre-trained VGG16 with feature extraction and fine-tuning.

- Evaluated performance using Accuracy, Loss, Confusion Matrix, and training curves.
- GitHub: Project Repository

Mushroom Classification with Naive Bayes & PCA (Individual Project)

- Built a machine learning pipeline for binary classification (edible vs poisonous) on a categorical dataset.
- Implemented Naive Bayes classifier from scratch with Laplace smoothing.
- Applied feature engineering and PCA for dimensionality reduction.
- Achieved up to 95
- GitHub: Project Repository

TECHNICAL SKILLS

Programming Languages

- Python, Java, C++, JavaScript, SQL

Machine Learning & Deep Learning

- Scikit-learn, TensorFlow, Keras
- CNNs, LSTMs, Transfer Learning
- Model Training, Evaluation, Hyperparameter Tuning

Natural Language Processing (NLP)

- NLTK, Hugging Face Transformers, LangChain
- Retrieval-Augmented Generation (RAG), FAISS
- Text Preprocessing, Tokenization, Lemmatization, TF-IDF, Word Embeddings

Data Analysis

- Pandas, NumPy, Matplotlib, Seaborn
- Data Cleaning, Feature Engineering, Jupyter Notebook

Data Engineering

- PySpark, Azure Data Factory, Databricks
- ETL Pipelines, Data Processing

Databases

- MySQL, Database Design

Tools

- Git, GitHub, Microsoft Azure, Streamlit

Soft Skills

- Problem Solving
- Teamwork
- Communication

EDUCATION

Faculty of Computers and Artificial Intelligence, Cairo University
Bachelor of Computer Science and Artificial Intelligence

2023 – 2027

COURSES

Data Analysis Course

IEEE

Sep 2024

Data cleaning, data visualization, and basic data analysis techniques.

Languages

- Arabic — Native
- English — Intermediate